A Physics-Informed Convolutional Neural Network for Predicting Clear-Air Turbulence Intensity (EDR) from ERA5 Reanalysis and Satellite Observations

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June 24, 2026

Abstract

Clear-air turbulence (CAT) is a growing and notoriously hard-to-predict aviation hazard: it occurs without visible cloud cues, injures hundreds of passengers and crew annually, and is projected to intensify as the jet stream strengthens under climate change. Operational nowcasting relies on systems such as NOAA's Graphical Turbulence Guidance (GTG), which combine many physical turbulence diagnostics through a *fixed weighted sum* calibrated against categorical pilot reports; most machine-learning alternatives, conversely, discard physics entirely. We present a *physics-informed* convolutional neural network (CNN) that bridges the two. From a purpose-built catalogue of 150 CAT events (2005–2024) we pull ERA5 reanalysis, GOES satellite brightness temperature, and in-situ ACARS eddy-dissipation-rate (EDR) measurements; we compute twelve operational CAT diagnostics (Richardson number, vertical wind shear, Ellrod indices, frontogenesis, etc.) on three jet-stream pressure levels; and we train an encoder–decoder CNN that emits a spatial EDR field, soft-aggregated to the scalar EDR the labels provide. The loss embeds physical constraints (non-negativity, shear–stratification gating, diagnostic consistency, spatial coherence, climatological bounds). We benchmark against a GTG-style physics surrogate and a gradient-boosted (XGBoost) baseline, and use an ablatable satellite stream to test whether cloud-top data adds skill over reanalysis alone. Six exploratory analyses—seasonal/diurnal decomposition, EMD and ensemble EMD, climate-teleconnection surrogate testing, regional ADS-B heatmaps, satellite–EDR cross-correlation, and PCA—motivate every design choice. A preliminary pure-NumPy CNN baseline attains an overall Pearson $r = 0.885$ (RMSE $0.120\ \mathrm{m}^{2/3}\,\mathrm{s}^{-1}$) but overfits (train $r = 0.933$ vs. validation $r = 0.736$) and is weak in the severe tail, directly motivating the regularised, physics-informed architecture. Rigorous validation of the full model (RQ1/RQ2) is the immediate next step.

1 Introduction

1.1 Why CAT matters

Clear-air turbulence is turbulence in the absence of convective clouds, typically in the strongly sheared flow near upper-tropospheric jet streams. Because it gives no visual or conventional radar warning, it is a leading cause of in-flight injuries. The May 2024 Singapore Airlines SQ321 event—a sudden severe encounter that caused a fatality and dozens of injuries—is a vivid reminder of the hazard. Beyond the immediate safety cost, several studies project that CAT frequency and intensity over busy mid-latitude corridors will rise through the century as lower-stratospheric warming sharpens the meridional temperature gradient and strengthens vertical wind shear [7, 8].

1.2 The prediction problem

We target the *eddy dissipation rate* (EDR), the ICAO-standard, aircraft-independent turbulence metric (units $\mathrm{m}^{2/3}\,\mathrm{s}^{-1}$). EDR is the rate at which turbulent kinetic energy cascades to ever-smaller eddies before dissipating as heat; it is estimated operationally from quick-access-recorder (QAR) accelerations and winds [2, 6]. EDR is heavy-tailed and approximately log-normal, so the rare, safety-critical severe tail is easily swamped in a linear loss. The clear-air setting compounds the difficulty: there is no cloud signature for an imager to key on, so prediction must come from the *dynamical state of the atmosphere*.

1.3 What is missing

The dominant operational tool, NOAA GTG/GTCC, is *not* a learning system: it computes $\sim$10 upper-level and $\sim$9 mid-level turbulence diagnostics per grid point and forms a fixed weighted sum, $\mathrm{EDR} = \sum_i w_i e_i$, with weights tuned against categorical PIREPs [3]. It ignores terrain and does not ingest satellite cloud-top information, which carries visual precursors of turbulence (transverse bands, billows, comma clouds). Pure-ML approaches, in turn, often abandon the hard-won physics of CAT diagnostics altogether.

1.4 Contributions and research questions

We build a physics-informed CNN that keeps the GTG diagnostic "vocabulary" but learns a *non-linear, spatial* combination of it, trained directly on continuous in-situ EDR rather than categorical reports. Our questions are:

- **RQ1.** Does a physics-informed CNN over reanalysis diagnostics predict in-situ EDR with skill beyond (a) a GTG-style physics surrogate and (b) a pure-ML XGBoost baseline?
- **RQ2.** Does satellite cloud-top brightness temperature add skill over reanalysis alone? We answer this structurally with an ablatable satellite stream.
- **Open question.** Should the model predict EDR *directly* or as a *physics residual*, EDR = surrogate + ML? We retain this as an explicit A/B for future work.

2 Background and Related Work

2.1 CAT physics and mechanisms

The leading CAT mechanism is Kelvin–Helmholtz (KH) instability in stratified shear flow. Where vertical wind shear is strong enough and static stability weak enough—quantified by a low Richardson number—small wave-like perturbations extract energy from the mean flow, grow, and break down into turbulence [9]. CAT is fundamentally a *mesoscale* phenomenon: the relevant structures span $\sim$10–100 km and observed CAT patches are typically less than $\sim$20 miles across, smaller than the synoptic patterns often used to forecast them. Secondary sources include orographic lee waves (terrain-forced gravity waves) and, near deep convection, convectively induced turbulence (CIT).

2.2 EDR as the turbulence metric

EDR has replaced aircraft-specific load metrics as the WMO/ICAO turbulence standard because it is intrinsic to the atmosphere rather than to a particular airframe [6]. It can be derived from vertical acceleration or from wind-based methods on QAR data [2], and is commonly mapped through a log-normal transform [5]. We adopt the standard severity bands used throughout: light ≥ 0.10, moderate ≥ 0.20, severe $\geq 0.40\ \mathrm{m}^{2/3}\,\mathrm{s}^{-1}$.

2.3 Operational nowcasting: NOAA GTG/GTCC

GTG ingests hourly RUC/RAP model output and computes a battery of diagnostics at each grid point, combining them as a weighted sum whose weights are optimised against PIREPs [3]. Our physics surrogate (§5) deliberately mirrors this design but improves on it in three ways drawn from the literature: it optimises an AUC objective rather than ψ, it regresses continuous in-situ EDR instead of categorical PIREPs, and it uses a log-normal diagnostic-to-EDR mapping [5].

2.4 Machine learning for turbulence

Prior ML pipelines typically preprocess flight data, window it, and run parallel regression (EDR) and classification (severity) heads [1]. Recurrent models (LSTMs), reduced-order/Koopman methods, and counterfactual analyses have all been proposed; we considered these but found a spatially-aware CNN better matched to the mesoscale structure of CAT and to our gridded reanalysis inputs.

2.5 Satellite signatures of turbulence

Operational forecasters use IR/visible cloud morphology—transverse bands, billows, comma clouds, delta cirrus, cyclonic curvature, lee slopes—as turbulence precursors. This motivates a satellite stream, while recognising (§4.5) that pure clear-air jet CAT has no cloud signature at all.

3 Data

All data sources are open-access and pulled programmatically. A Google Cloud `e2-standard-2` VM (2 vCPU, 8 GB; Intel Broadwell; accessed over SSH) handled the heavier acquisition jobs.

3.1 Acquisition pipeline

The pipeline is five staged scripts (`scripts/step1--4`, `fetch_opensky.py`):

1. **Event discovery** (`step1`): scan 20 years of MADIS/ACARS reports at 3-hourly cadence (00,03,…,21 UTC), filter to FL180+, and cluster reports spatially (200 km) and temporally (6 h) via a pure-Python $O(n^2)$ routine with a 6-hour early-break. Clusters are stratified into severity bins, expanding an initial hand-picked set of 10 events to 150.

2. **Per-event ACARS** (`step2`): download the full ±window of ACARS reports (turbulent and smooth) per event.

3. **Per-event ERA5** (`step3`): pull the reanalysis fields below from ARCO-ERA5.

4. **Per-event satellite** (`step4`): pull GOES IR brightness-temperature statistics.

5. **Context** (`fetch_opensky`): OpenSky ADS-B state vectors for the regional analysis.

Table 1: Representative rows from `events.csv`.

ID	start (UTC)	loc.	max	mean	bin
149	2007-06-10	42N 91W	0.15	0.15	light
84	2006-04-14	43N 78W	0.25	0.15	moderate
40	2006-03-12	43N 90W	0.35	0.17	moderate
66	2024-02-12	35N 86W	0.26	0.15	moderate
0	–	–	0.95	–	severe

3.2 Event catalogue

The labelled unit of analysis is an "event." The catalogue (`events.csv`) contains **150 events** spanning 2005-10-07 to 2024-12-30 (mean duration $\approx$4.6 h). Each row records the centroid, start/end time, bounding box (centroid $\pm 2^\circ$), report count, severity bin, and the scalar labels `max_edr` and `mean_edr` (Table 1). The label distribution is `max_edr` min 0.150 / mean 0.366 / max 0.950, with bins light 37 / moderate 76 / severe 37. **Crucially, the labels are scalar**—no gridded EDR exists—which drives the soft-aggregation design in §5.

3.3 ACARS / MADIS in-situ EDR (labels)

Ground truth comes from the NOAA MADIS public archive, providing per-report median (`MEDEDR`) and maximum (`MAXEDR`) EDR, position, altitude, time, and tail number, plus vertical acceleration, peak vertical gust, and a turbulence index. We retain cruise-altitude reports ($\geq$ 5500 m, FL180+) sampled 3-hourly; both turbulent and smooth reports are kept per event.

3.4 ERA5 reanalysis (physics fields)

We pull ERA5 from the analysis-ready ARCO-ERA5 Zarr store on Google Cloud (`gcp-public-data-arco-era5`; anonymous, no API key), at 0.25°, hourly, averaged over a $\pm$2 h window around each event. Variables: u, v (wind), T (temperature), z (geopotential), ω (vertical velocity), q (specific humidity). Pressure levels 150, 175, 200, 225, 250, 300, 350, 400, 500 hPa are requested and filtered to those present; the primary jet band is 225/250/300 hPa, with flanking levels supporting vertical derivatives.

3.5 Satellite: GOES cloud-top brightness temperature

We use GOES-16/19 ABI Band 13 (10.3 µm IR window) from the NOAA AWS buckets (`s3://noaa-goes16/19`, 2017+, with legacy GOES-13 for 2010–2017). Rather than gridded imagery we store *scalar per-snapshot statistics* over the event box: `tb_min`, `tb_max`, `tb_mean`, `tb_std`, plus scan time, source path, and satellite ID. **Coverage caveat central to RQ2:** only **64 of 150 events** have satellite data, all post-2017; severe pre-2017 jet CAT has no cloud signal, so the satellite stream must be structurally optional.

3.6 Supplementary sources

COSMIC-2 GNSS radio-occultation profiles (refractivity, temperature, pressure vs. geopotential height) provide extra vertical structure. OpenSky ADS-B state vectors (2022 sample, FL180+ cruise) supply the regional turbulence-proxy analysis of §4.4 but are not model inputs. The NTSB aviation query aided event discovery.

3.7 Software and reproducibility

Python 3.11 with `xarray/cfgrib/netCDF4`, `pandas/numpy/pyarrow`, `metpy`, `scipy`, `scikit-learn`, `xgboost`, `torch`, `optuna`, `s3fs/satpy/goes2go`, and `matplotlib/cartopy`. Diagnostics are cached per event; all code is open-sourced.

4 Exploratory Data Analysis

Before building the model we examined the dataset for shared structure and trends, starting from the most prevalent events. Six analyses each motivate a concrete modelling choice; we present each as method → finding → implication.

4.1 Seasonal and diurnal climatology

Over **116,245,659** ACARS reports (2004–2024, FL180+; anomalous years 2009–2011 excluded for event-window sampling bias, see below), turbulence peaks in **January** (0.009% of reports moderate) and troughs in **August** (0.003%); the seasonal cycle peaks near day-of-year 252 and troughs near DOY 353, a winter-dominant pattern set by the polar jet (Fig. 1a). The diurnal cycle is weak (peak 21 UTC, amplitude only 0.052 pp, Fig. 1b), implying the turbulence here is **mechanical/jet-driven, not convective**: convective CAT would show a strong afternoon peak. Turbulence fraction rises with altitude, peaking in the FL180–260 band (Fig. 1c), consistent with maximum shear just below the winter tropopause. A linear trend on clean years is slightly *decreasing* (moderate -0.006 pp/decade, p=0.017; severe -0.002 pp/decade, p=0.032; Fig. 1d), opposite to North Atlantic studies—likely a narrower window, North-American routes, and a fleet-composition caveat.

A seasonal–trend decomposition (Fig. 2) partitions daily-scale variance as **4.8% seasonal, 5.4% trend, and 89.8% residual (synoptic)**. **This ∼90%-synoptic result is the central justification for the model design**: day-to-day CAT is governed by individual storms and jet meanders, so ERA5 physics fields must do the heavy lifting and season/climate enter only as weak priors. Accordingly we encode date/time as cyclic *FiLM conditioning* (§5), letting the network decide how much to use them, and keep the diurnal term minimal. The month×hour heatmap (Fig. 3) makes the dominance of the seasonal over the diurnal axis visually explicit.

Data-quality note. Years 2009–2011 show `frac_m` ∼100× above neighbours while flight volume only doubled—an *event-window sampling bias* from the seeded download. They are excluded from all climatology/trend statistics and interpolated in the decomposition.

4.2 Spectral decomposition: EMD and EEMD

We decomposed the monthly turbulence record into intrinsic mode functions (IMFs) using Empirical Mode Decomposition. Because the metric choice matters, we ran two complementary analyses on two signals.

EMD on occurrence rate (Fig. 4). On the monthly fraction of reports exceeding moderate (`frac_m`), single-trial EMD finds that the *sub-seasonal* band dominates: IMF1 (∼4 mo) carries ≈37.5% of variance and IMF2 (∼9 mo) ≈33.9%, with *no clean 12-month mode*. The interpretation is decisive for modelling: **month-to-month turbulence rate is driven by transient synoptic**

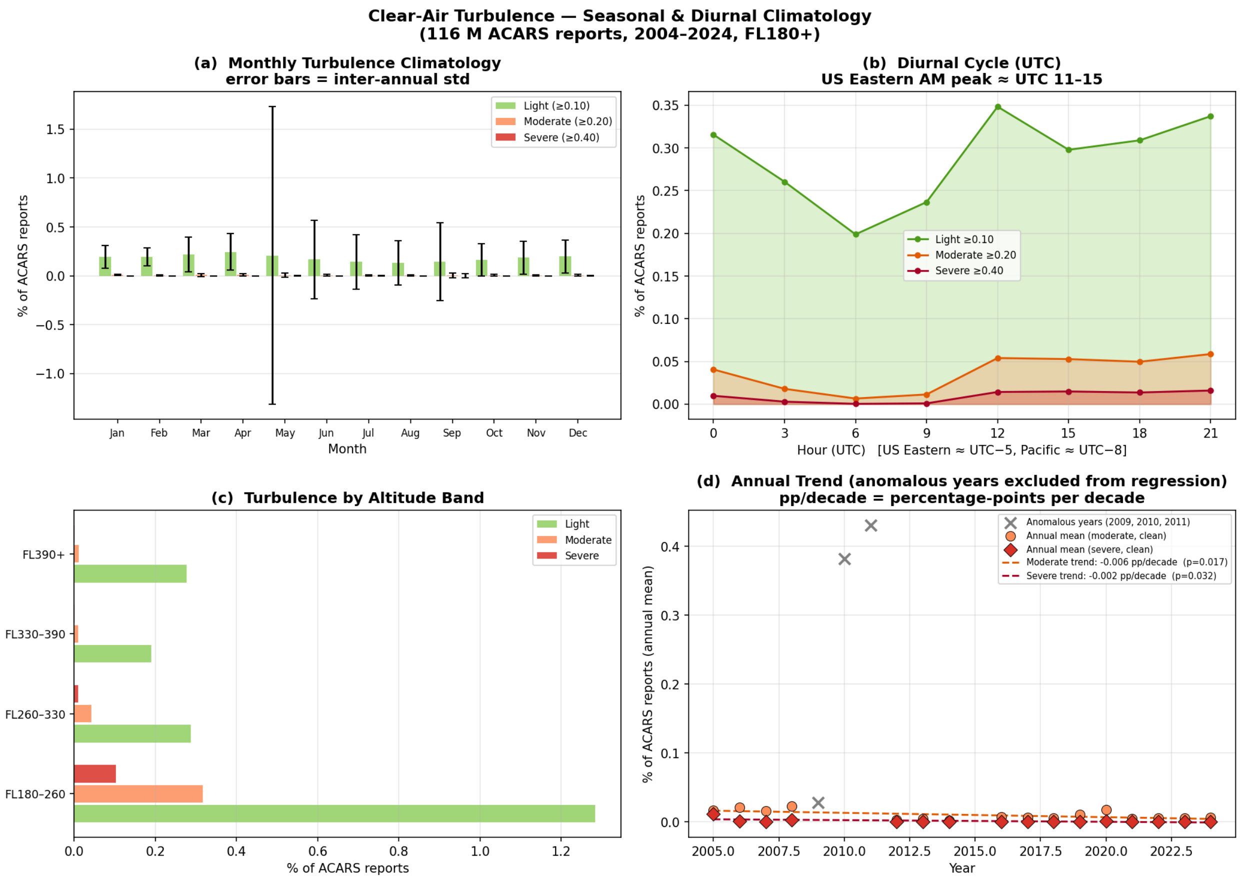

Figure 1: Seasonal and diurnal climatology of ACARS turbulence (116 M reports, FL180+). (a) Monthly climatology with inter-annual error bars (January peak); (b) weak diurnal cycle (21 UTC peak, amplitude 0.052 pp); (c) turbulence by altitude band (peak FL180–260); (d) slightly decreasing annual trend, anomalous years (grey $\times$) excluded from the regression.

weather, not the calendar—so date alone cannot predict events, and synoptic ERA5 fields are essential.

EEMD on intensity (Fig. 5). Plain EMD suffers *mode mixing*; Ensemble EMD (EEMD; 100 noise-perturbed trials, noise width 0.05) averages it away for cleaner timescale separation. Applied to the monthly mean above-threshold intensity (`edr_mean_above`; mean 0.269 $\mathrm{m}^{2/3}\,\mathrm{s}^{-1}$), EEMD reveals substantially more *low-frequency* energy: multi-year ($\sim$4 yr, 11.7%) and decadal ($\sim$8 yr, 18.8%) modes are prominent. **It is this low-frequency intensity energy that prompted the climate-teleconnection test** of §4.3. Per-IMF percentages are method-sensitive (IMFs may be mutually correlated and need not sum to 100%), so we treat them as indicative and lead with the robust 89.8% synoptic figure of §4.1.

4.3 Climate-index teleconnections

We matched each EEMD mode to the climate index nearest its native band (ONI/ENSO, Niño3.4, QBO, PDO, NAO, AO, MJO) and computed lagged cross-correlations over $\pm$12 months on 170 valid months. Because two narrowband series correlate by chance, significance was assessed against **1000 Fourier phase-randomised surrogates**, and a Hilbert amplitude-envelope test asked whether

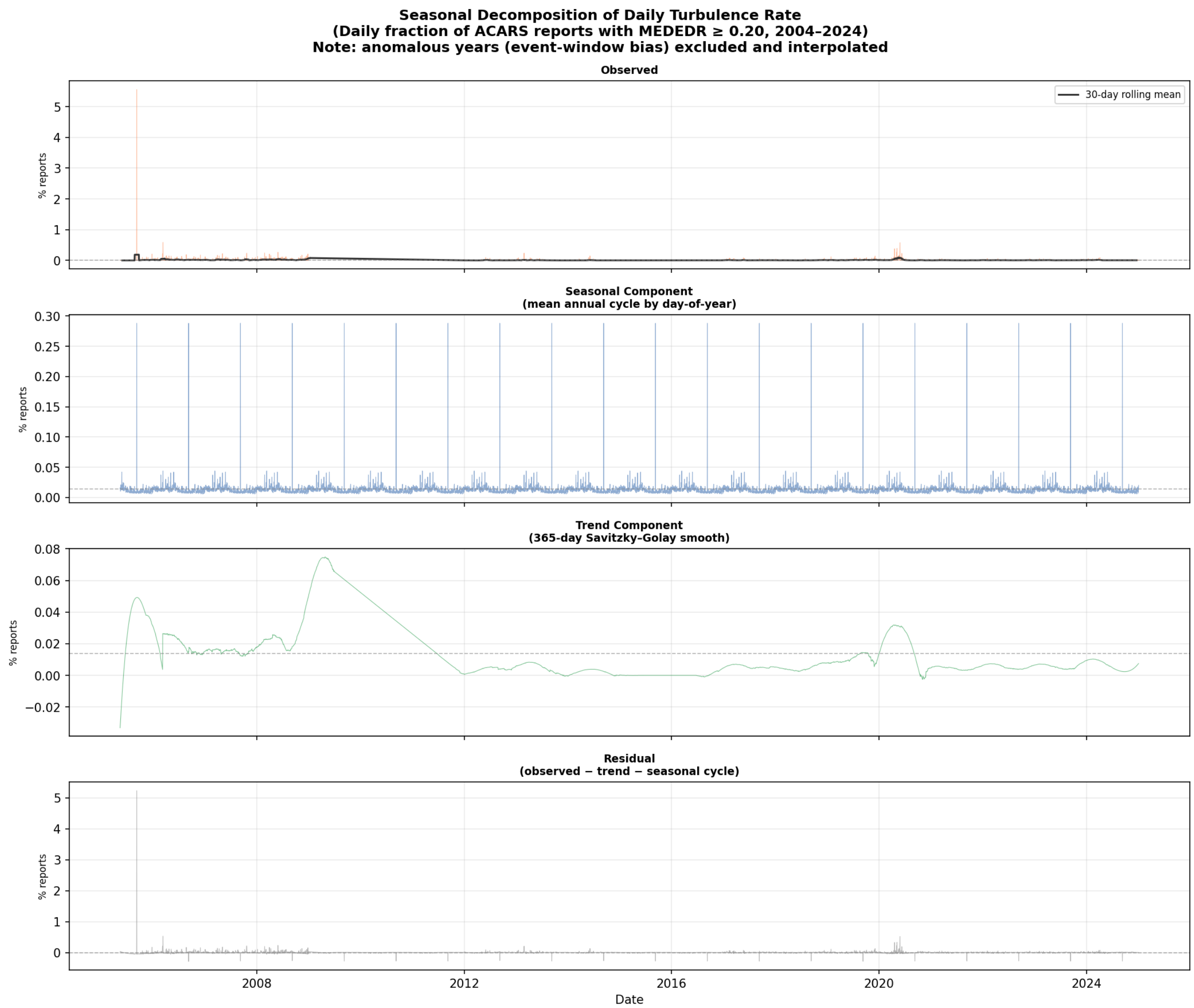

Figure 2: Seasonal decomposition of the daily turbulence rate (observed, seasonal, 365-day Savitzky–Golay trend, residual). Variance partitions as seasonal 4.8%, trend 5.4%, and residual (synoptic weather) **89.8%**—the result that motivates a physics-field-driven model.

turbulence variability *strengthens* in a given climate phase (Fig. 7). Phase correlations are modest ($|r| = 0.14$–0.42) and **none are significant** ($p > 0.05$) under surrogates; large envelope correlations (e.g. PDO +0.98) are inflated by autocorrelation and too few independent cycles in a 20-year record. Lead/lag signs are mostly *negative* (turbulence leading the index), the signature of a shared trend rather than a predictive teleconnection.

Conclusion: there is no statistically robust monthly climate teleconnection for North American CAT. **Implication:** we include ONI, Niño3.4, PDO, and QBO as model inputs but impose *no* down-weighting—the network learns their (small) weight. A winter-stratified NAO/AO analysis is left to future work.

4.4 Regional spatial distribution (ADS-B)

Using OpenSky ADS-B state vectors we built a turbulence proxy—intra-hour standard deviation of vertical rate $> 1.5\,\mathrm{m\,s^{-1}}$, calibrated to MEDEDR ≥ 0.20 following Kim et al. [2], Sharman and

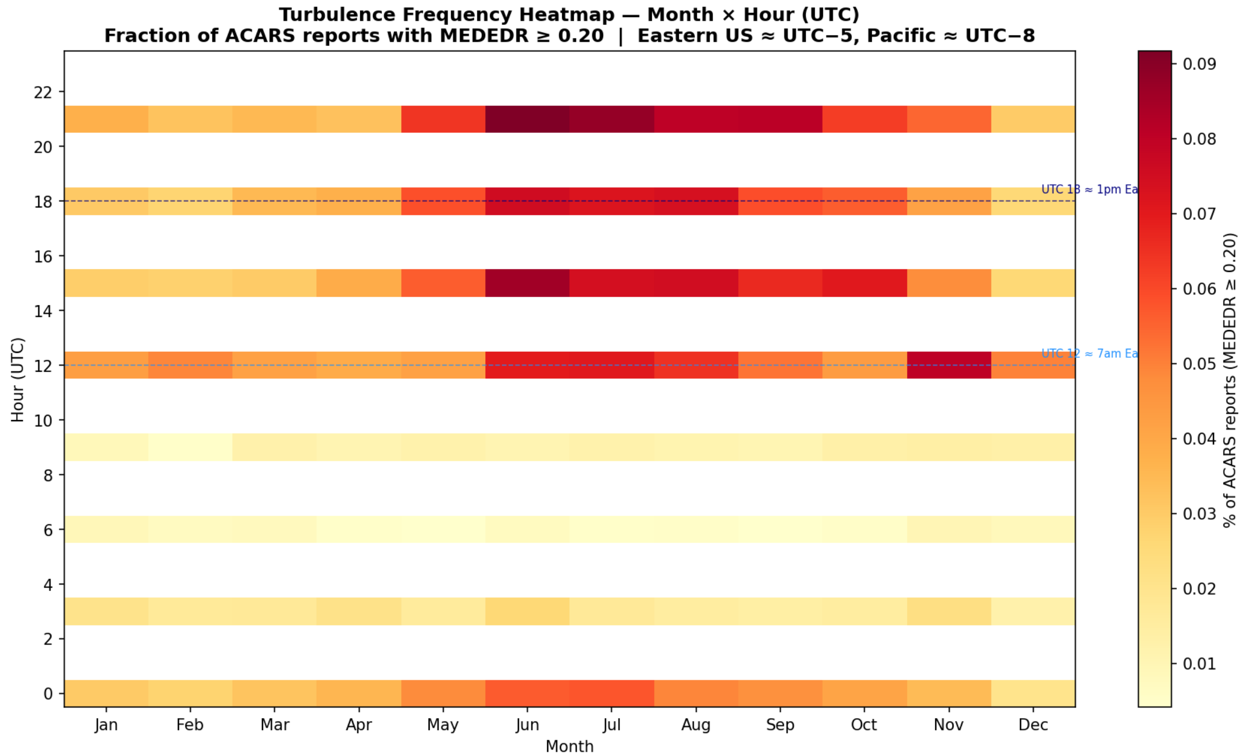

Figure 3: Month × hour (UTC) turbulence-frequency heatmap. The seasonal axis dominates the (very weak) diurnal axis, consistent with jet-stream rather than convective forcing.

Lane [4]—and mapped month × hour encounter rates across 15 global aviation regions (dateline-aware; FL180+ cruise; ≥5 obs/aircraft-hour; Fig. 8). The clear geographic and seasonal structure of encounter rates supports a spatial (CNN) treatment and the box-based event sampling we adopt.

4.5 Satellite TB ↔ EDR cross-correlation

On the 64 satellite-covered events (431 lag-0 pairs), Spearman correlations against hourly mean EDR are: `tb_mean` $r = -0.27$ ($p < 0.001$), `tb_max` $r = -0.20$ ($p < 0.001$), `tb_min` $r = -0.09$ (n.s.), `tb_std` ≈ 0 (n.s.). The best *forward* predictor is `tb_mean` at $+2\,$h ($r = -0.31$): colder tops precede turbulence, suggesting a ∼2-hour lead (Figs. 9–11). The off-trend outliers in the scatter (Fig. 10) are precisely the severe clear-sky events with no cold cloud—the signal is driven by post-2017 moderate/convective cases and is **blind to severe jet CAT**.

Implications: the satellite enters as an *optional, droppable* stream with an explicit present/absent mask, with suggested encodings `tb_std`, $(220 - \texttt{tb\_min})$, and a cooling Δ`tb`. This analysis directly frames **RQ2**.

4.6 Principal component analysis

PCA of 16 ACARS event features shows PC1 (32.8% variance) is an EDR-intensity axis and PC2 (23.2%) contrasts vertical extent with geographic spread; severe events separate cleanly along +PC1, and the first three PCs explain >80% of variance (Fig. 12). PCA of 29 ERA5 features yields a PC1 (41.4%) that opposes vertical velocity (ω) against per-level wind speed—a shear/stability axis—but events coloured by max EDR show only *weak* separation (Fig. 13). This is itself a design argument:

Table 2: Physics diagnostics computed per pressure level (`src/features/diagnostics.py`). θ is potential temperature; ζ relative vorticity; $\mathbf{V}$ horizontal wind.

Diagnostic	Formula	Role
VWS	$\sqrt{(\partial_z u)^2 + (\partial_z v)^2}$	KH energy source
N^2	$(g/\theta)\,\partial_z\theta$	stratification
Ri	N^2/VWS^2	KH instability (< 0.25)
DEF	$\sqrt{\mathrm{DST}^2 + \mathrm{DSH}^2}$	frontal tightening
DIV	$\partial_x u + \partial_y v$	convergence
TI1	$\mathrm{VWS} \cdot \mathrm{DEF}$	Ellrod index 1
TI2	$\mathrm{VWS}(\mathrm{DEF} - \mathrm{DIV})$	Ellrod index 2
ζ	$\partial_x v - \partial_y u$	rotation
FRONTO	Petterssen frontogenesis	jet/front CAT
WSPD	$\sqrt{u^2 + v^2}$	jet-core proxy
ω	(raw ERA5)	GW/convective forcing
VADV	$-\mathbf{V} \cdot \nabla\zeta$	NVA predictor

bulk atmospheric state is *not* linearly separable by severity, so **spatial structure (a CNN) is needed** to extract skill that pooled features miss—and it foreshadows the XGBoost-vs-CNN test of §6. Geographic projection of the scores shows mild clustering, with the highest-PC1 (most intense) events at higher latitudes (Fig. 14).

5 Methodology

The headline model is a physics-informed CNN; a GTG-style surrogate and an XGBoost regressor serve as baselines, and a pure-NumPy CNN provides a simple reference. The pipeline is: ERA5 + satellite + ACARS $\to$ diagnostics $\to$ CNN $\to$ EDR field $\to$ soft-aggregated scalar EDR, supervised on `max_edr`. The model inherits GTG's diagnostic vocabulary but replaces its fixed weighted sum with a learned, spatial, non-linear mapping.

5.1 Physics diagnostics (inputs)

Rather than feeding raw u, v, T, we compute 12 operational CAT diagnostics per pressure level on the native ERA5 grid, then bilinearly resize to 24×24; heavy-tailed channels ({VWS, DEF, TI1, TI2, WSPD}) are log-compressed before z-scoring, and all diagnostics are cached per event. Table 2 lists them. With three primary levels (225/250/300 hPa) this yields $12 \times 3 =$ **36 input channels**. Constants: g_0=9.80665, R_d/c_p=0.286, P_0=1000 hPa; Ri is clipped to $[-5, 50]$ to tame the $1/\mathrm{VWS}^2$ tail.

5.2 Auxiliary input streams

Beyond the 36 diagnostic channels: four **climate** scalars (ONI, Niño3.4, PDO, QBO from NOAA PSL; broadcast as channels and/or via FiLM); four **cyclic-time** features (sin / cos of DOY and hour, keeping Dec/Jan adjacent); five **satellite** features (`tb_cold`, `tb_mean`, `tb_std`, `tb_max`, cooling Δ`tb`) plus a present/absent mask; and three raw physics fields (Ri, VWS, TI1) passed un-normalised for the physics losses. Per-channel z-scoring is fit on the training split only and persisted to `*_norm.npz`.

5.3 Reference model: pure-NumPy CNN

A from-scratch NumPy CNN (`src/models/cnn_cat.py`) provides a simple, dependency-free reference. It takes 18 raw ERA5 channels (6 variables $\times$ 3 levels) on a 24×24 grid through Conv(18$\rightarrow$16,k3)$\rightarrow$ReLU$\rightarrow$MaxPool$\rightarrow$Conv(16$\rightarrow$32)$\rightarrow$ReLU$\rightarrow$MaxPool$\rightarrow$GlobalAvgPool$\rightarrow$FC(32$\rightarrow$32)$\rightarrow$ReLU$\rightarrow$FC 1)$\rightarrow$sigmoid ($\sim$8.3k parameters), regressing `max_edr`/0.95 under MSE with Adam. Its preliminary results (§6) define the behaviour the physics-informed model must improve on.

5.4 Physics-informed CNN

The headline network (`src/models/cnn_torch.py`) is a two-stream encoder–decoder (U-Net-like) with `same` padding throughout (default depth 3: 24$\rightarrow$12$\rightarrow$6$\rightarrow$3; base width 32 doubling per block; Conv$\rightarrow$Norm$\rightarrow$Act$\rightarrow$Dropout2d$\rightarrow$Conv$\rightarrow$Norm$\rightarrow$Act). *FiLM conditioning* after each encoder block applies per-channel $\gamma \odot x + \beta$ generated from the time (and optionally climate) vector, identity-initialised so conditioning starts as a no-op. A small *satellite MLP* produces an embedding fused at the bottleneck and is **ablatable**: when satellite is absent the input is zeros plus a mask, so the model learns to ignore it—this toggle is the structural test for RQ2. The decoder upsamples with skip concatenation; a 1×1-convolution *FNN head* (a shared per-pixel MLP) maps decoder features to the field, and a **softplus** output enforces EDR ≥ 0 without saturating the heavy tail.

Because labels are scalar, a differentiable **soft aggregation** reduces the predicted field to a scalar: log-sum-exp with searchable temperature τ (default 8.0), or a top-k mean. The spatial mean is aggregated separately to an auxiliary `mean_edr` target. The model has $\sim$80k parameters.

5.5 Loss: log target plus physics penalties

We train on $\log 1\mathrm{p}(\mathrm{EDR})$ (inverting with `expm1`) to stabilise the heavy tail—a deliberate change from the linear/sigmoid target of the NumPy reference. The base loss is a magnitude-weighted Huber (δ=0.1) on $\log 1\mathrm{p}(\hat{\mathrm{EDR}}_{\mathrm{max}})$ vs. $\log 1\mathrm{p}($`max_edr`$)$, with weight $1 + w_{\mathrm{mag}} y$ emphasising the rare severe tail, plus an auxiliary Huber on the mean (w_{aux}=0.2). Five physics-penalty terms, each with a searched weight λ_i, shape the *field*:

1. **Non-negativity** backstop, $\mathrm{relu}(-\hat{y})^2$.
2. **Shear–stratification gating**: penalise high predicted EDR where flow is dynamically stable ($\mathrm{Ri} \gg 1$) *and* low-shear—no available energy (encodes KH theory; $\mathrm{Ri}_{\mathrm{crit}}$=1, gate sharpness 2).
3. **Ellrod-TI consistency**: penalise negative rank-correlation between the predicted field and the TI1 field (the CNN may refine but should not invert the physics).
4. **Total variation**: a small penalty enforcing the coherent patch structure of CAT.
5. **Climatological bound**: penalise predictions exceeding the observed maximum (0.95).

5.6 Baselines and the XGBoost comparison

Physics surrogate (Model 1, GTG/GTCC-style). Log-normal-calibrate each diagnostic to an EDR estimate, then optimise a weight vector for ROC-AUC at the moderate threshold via differential evolution, comparing the fitted weights to GTG climatological weights as a sanity check. *(Specified; build is partial.)*

XGBoost (Model 2, pure ML). The gradient-boosted baseline (`src/models/baseline_xgb.py`) regresses log 1p(`max_edr`) on **spatially pooled** features: the per-channel mean, max, and standard deviation of all 36 diagnostics, concatenated with the climate, cyclic-time, and satellite scalars ($\sim$121 features). Hyperparameters are deliberately modest (400 trees, `max_depth` 4, learning rate 0.03, subsample/colsample 0.8) and it is evaluated out-of-fold.

Why XGBoost is the decisive comparison. XGBoost collapses each event's spatial fields to summary statistics, so it captures *how strong* the diagnostics are but discards *where* and *in what spatial arrangement*. The CNN, by contrast, sees the full 24×24 structure. **The CNN-vs-XGBoost gap therefore isolates the value of spatial structure itself**: if the CNN does not beat XGBoost, the convolutional machinery is not earning its keep and pooled physics features suffice. The PCA of §4.6—where bulk ERA5 state separates severity only weakly—predicts that pooled features will plateau and that spatial context should help; XGBoost makes that hypothesis testable, and also supplies an interpretable feature-importance ranking to cross-check the CNN against the physics. XGBoost additionally serves as the substrate for the residual-hybrid open question (§5.8).

5.7 Training and model selection

We use a **temporal hold-out** (most-recent 20% of events) plus **event-grouped, severity-stratified *k*-fold** CV so no event leaks across the train/validation boundary—a stricter protocol than the simple 80/20 split used for the NumPy reference. Optimisation uses AdamW (lr 10^{-3}, weight decay 10^{-4}, batch 16), up to 200 epochs with patience 30 and gradient clipping at 5; augmentation is restricted to axis flips (wind vectors are not rotation-invariant). An **Optuna** study (TPE sampler, median pruner; 40 trials $\times$ 4-fold, 80-epoch budget) searches architecture (depth, width, kernel, norm, activation, pooling, dropout), FiLM, the satellite on/off toggle, the FNN head, aggregation type and τ/k, the optimiser, and all loss λ's. The search objective is log-RMSE+(1−AUPRC@0.20), balancing regression accuracy and rare-event discrimination.

5.8 Open design question: direct vs. residual

The architecture supports both direct prediction (built) and a *physics-residual* formulation EDR = surrogate(Ellrod/GTG) + ML(residual). We retain this as an explicit planned A/B rather than baking it in: residual learning can ease training on small data and gives a clean "does ML beat physics" narrative, while direct prediction avoids surrogate bias.

5.9 Metrics

We report RMSE and Pearson correlation on log-EDR; AUROC, AUPRC, CSI, and TSS for moderate exceedance (EDR $\geq$ 0.20); and the Brier score with a reliability diagram.

6 Results

6.1 Preliminary baseline (NumPy CNN)

The pure-NumPy reference (§5.3) has been trained on all 150 events under a simple 80/20 split; its event-level predictions (`models/cat_cnn_eval.csv`) give the first real signal that the problem is learnable. Table 3 and Fig. 15 summarise it.

Table 3: Preliminary NumPy-CNN results (18 raw ERA5 channels, simple 80/20 split, *not* the rigorous protocol of §5).

Split / bin	n	RMSE	MAE	Pearson
All	150	0.120	0.076	0.885
Train	121	0.089	0.061	0.933
Validation	29	0.202	0.137	0.736
Severe	37	0.199	0.137	0.669
Moderate	76	0.063	0.048	0.445
Light	37	0.102	0.072	−0.107

Table 4: Planned model comparison on the temporal hold-out. The NumPy row carries the preliminary numbers of Table 3 (different, non-grouped split—shown for context only).

Model	RMSE	Pearson	AUPRC@0.20
Ri-only baseline	—	—	—
Physics surrogate (GTG)	—	—	—
XGBoost (pooled)	—	—	—
NumPy CNN (prelim)	(0.120)	(0.885)	—
Physics CNN	—	—	—
+ satellite	—	—	—

Three findings carry straight into the design of the physics-informed model. **(i) The signal is real**: overall $r = 0.885$ shows ERA5 fields do encode EDR. **(ii) It overfits**: the train→validation drop (r 0.933→0.736; RMSE 0.089→0.202) on only 150 events demands the heavier regularisation (dropout, weight decay, physics penalties, augmentation) and the leakage-safe grouped CV we adopt. **(iii) The tail is hardest**: severe-bin RMSE (0.199) dwarfs moderate (0.063), and the light bin even has slightly negative correlation—exactly the heavy-tail pathology the log target and magnitude-weighted loss are designed to fix. These numbers are *not* comparable to the planned results below because they use raw channels and a non-grouped split; they are a sanity floor, not a headline.

6.2 Planned model comparison (RQ1)

The rigorous three-way comparison—physics surrogate vs. XGBoost vs. physics-informed CNN, on the temporal hold-out with grouped CV—is the core RQ1 experiment (Table 4). The CNN-vs-XGBoost row is the decisive test of whether spatial structure adds skill (§5.6). *[PENDING — validation]*

6.3 Satellite-skill ablation (RQ2)

ERA5-only vs. ERA5+satellite on the post-2017 subset, toggling the ablatable stream of §5. *[PENDING — validation]*

6.4 Further ablations

Direct vs. residual (§5.8); physics penalties on/off; and the CNN-vs-XGBoost spatial-structure test. *[PENDING — validation]*

6.5 Headline figures

Diagnostic sanity maps (low-Ri/high-TI1 co-locating with severe-event positions), predicted-vs-observed EDR maps for test events (e.g. SQ321), XGBoost feature importance, and ROC curves at EDR ≥ 0.20. *[PENDING — validation]*

7 Discussion

The preliminary baseline already tells a coherent story: the dynamical fields carry strong EDR signal ($r = 0.885$), but with 150 events the binding constraint is *generalisation*, not raw capacity—hence a design that injects physics as inductive bias (diagnostic inputs, KH gating, diagnostic-consistency and coherence penalties) rather than adding free parameters. The remaining quantitative discussion—skill relative to operational benchmarks (GTG, ECMWF IFS-CAT) and the interpretation of the satellite ablation against the coverage caveat of §4.5—awaits the rigorous run. *[PENDING — validation]*

Honest limitations. (i) ERA5's 0.25° resolution is coarse relative to the <20-mile CAT patch scale. (ii) Labels are *scalar only*: the predicted field's shape is physics-constrained, not label-supervised, until gridded EDR exists. (iii) Ground truth is sparse (150 events; satellite on only 64), and the preliminary baseline's overfitting shows how tight that is. (iv) Convective-vs-true-CAT ambiguity persists in the post-2017 satellite subset. (v) The pre-2017 satellite gap coincides with the most severe events. (vi) The decreasing-trend result may carry instrumental (fleet-drift) signal. (vii) EMD/EEMD per-IMF variance partitions are method- and signal-sensitive and are read qualitatively.

8 Conclusion and Future Work

We have assembled an open, 150-event, multi-source CAT dataset and built a physics-informed CNN that predicts EDR while respecting CAT physics through both its inputs (operational diagnostics) and its loss (KH gating, diagnostic consistency, spatial coherence, climatological bounds), with a GTG-style surrogate and XGBoost as baselines and an ablatable satellite stream to answer whether imagery adds skill. A preliminary NumPy reference confirms the signal is learnable ($r = 0.885$) while exposing the overfitting and heavy-tail behaviour the full design targets. Rigorous validation (RQ1/RQ2) is the immediate next step. Future work includes the direct-vs-residual A/B; promoting the heatmap to dense supervision via sparse-to-dense ACARS-track labels (changing only the loss); Himawari coverage for global regions; COSMIC-2 vertical structure; a winter-stratified NAO/AO teleconnection test; and a real-life test that simulates a weather phenomenon and predicts CAT hotspots.

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A Diagnostic equations (GTG cross-reference)

The diagnostics of Table 2 map to standard GTG equation labels: Ri (A1), structure-function EDR (A7) and SIGW (A8), frontogenesis (A9/A10), TI1 (A15), horizontal temperature gradient (A23), wind speed (A24), $|\mathbf{V}|\cdot$DEF (A28), unbalanced-flow diagnostic (A30), ABSIA (A34), NCSU1 (A36), Colson–Panofsky (A4), and DTF3 (A6). *[PENDING — full derivations]*

B Hyperparameters and search space

The complete fixed configuration and Optuna search space are in `configs/cnn.yaml`: defaults grid 24, levels 225/250/300, depth 3, base width 32, GroupNorm, GELU, max-pool, dropout 0.1, FiLM hidden 32, satellite MLP 16, FNN depth 2/width 64, log-sum-exp τ=8; loss δ=0.1, w_{mag}=1, w_{aux}=0.2, λ_{Ri}=0.1, λ_{TI}=0.05, λ_{TV}=0.01. *[PENDING — full table dump]*

Figure 4: EMD of the monthly turbulence *rate* (`frac_m`). Sub-seasonal IMF1 (~4 mo, 37.5%) and IMF2 (~9 mo, 33.9%) dominate; no clean annual mode appears, confirming that synoptic weather—not the calendar—drives monthly turbulence rate.

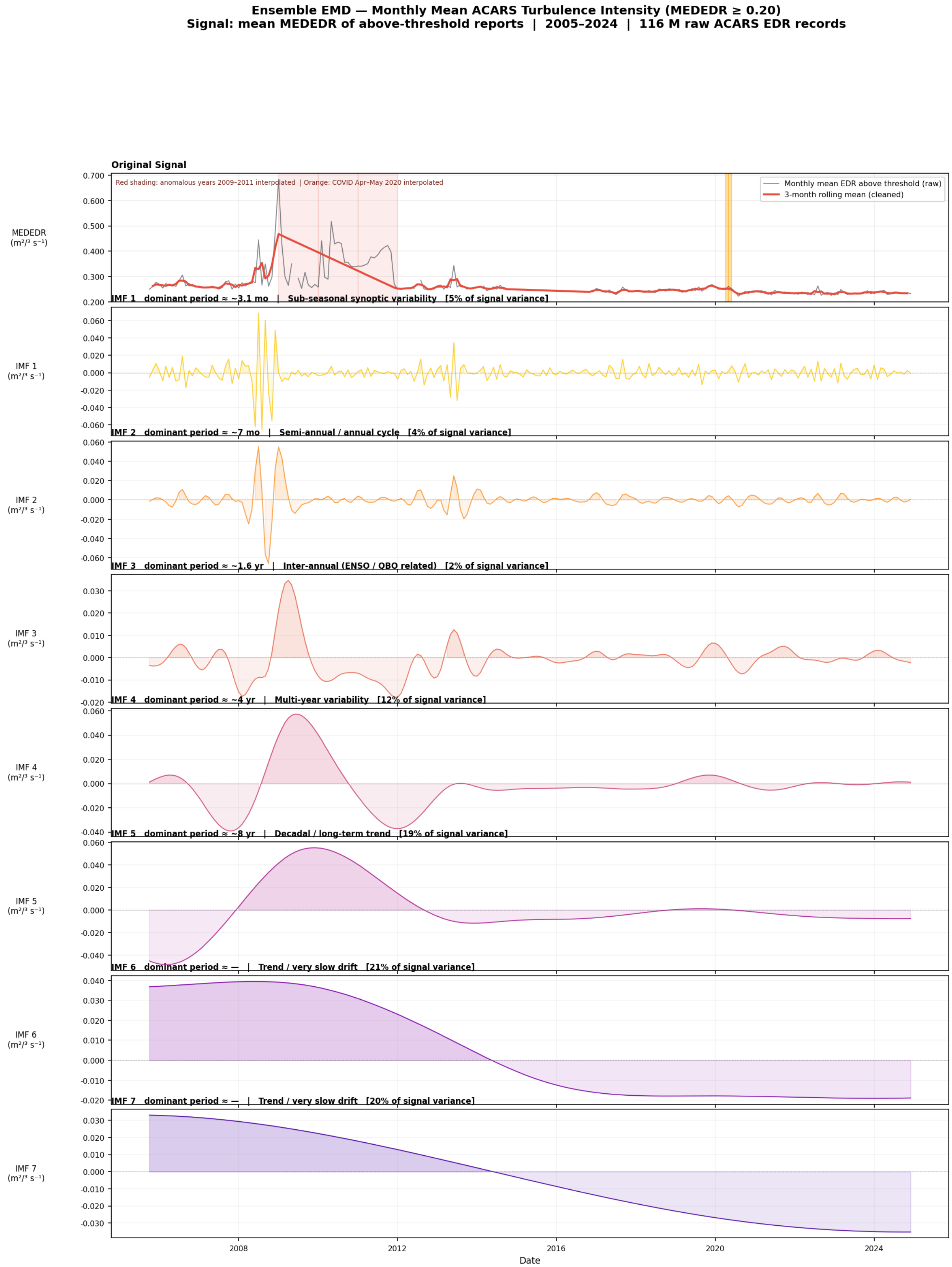

Figure 5: EEMD (100 trials) of the monthly turbulence *intensity* (`edr_mean_above`). Ensemble averaging cleanly separates sub-seasonal through decadal modes and exposes prominent multi-year (∼4 yr) and decadal (∼8 yr) energy that motivated the climate-teleconnection test (Fig. 6).

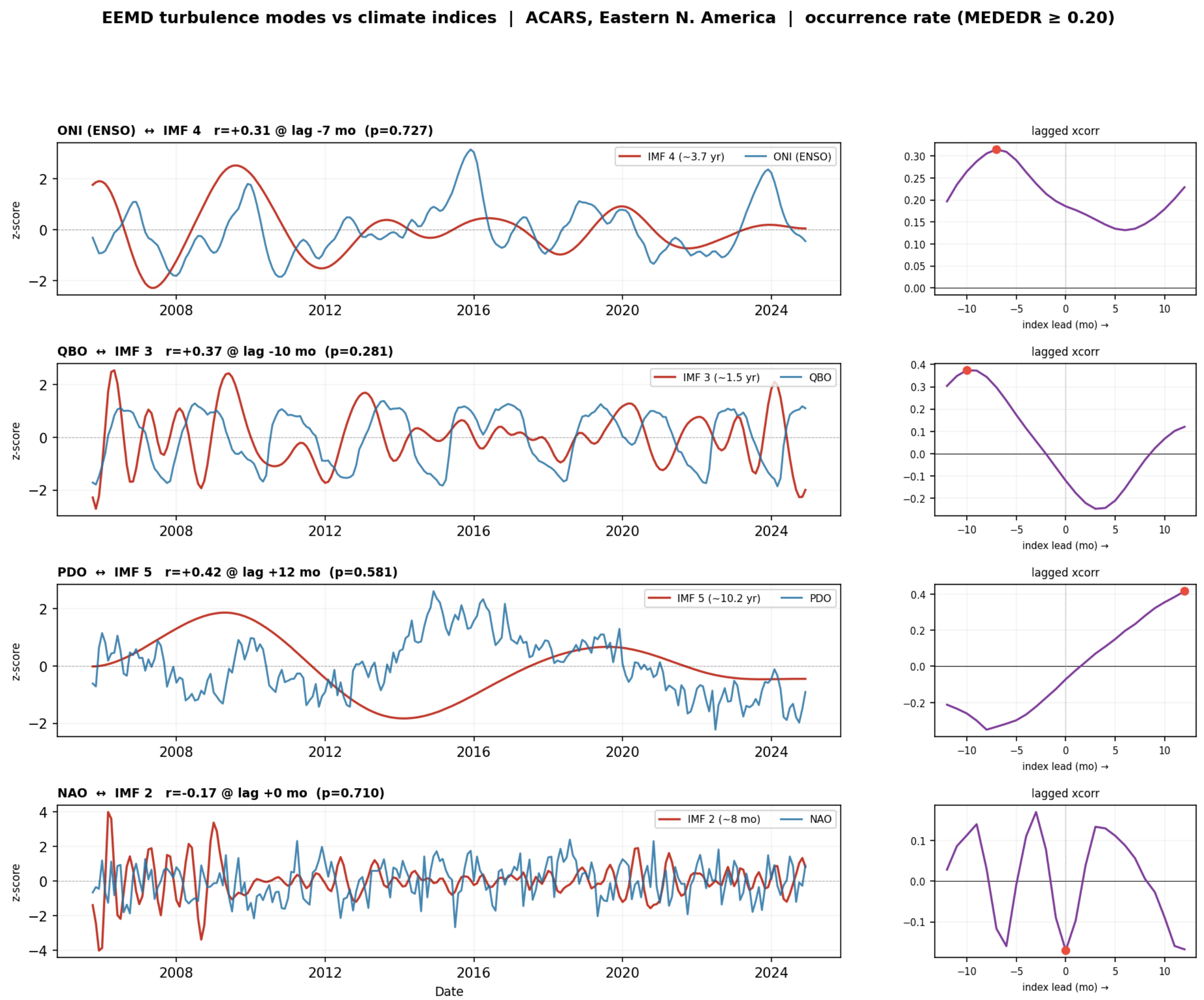

Figure 6: Lagged cross-correlations of EEMD turbulence modes against climate indices (ONI, QBO, PDO, NAO). All correlations are non-significant under 1000 phase-randomised surrogates ($p > 0.05$).

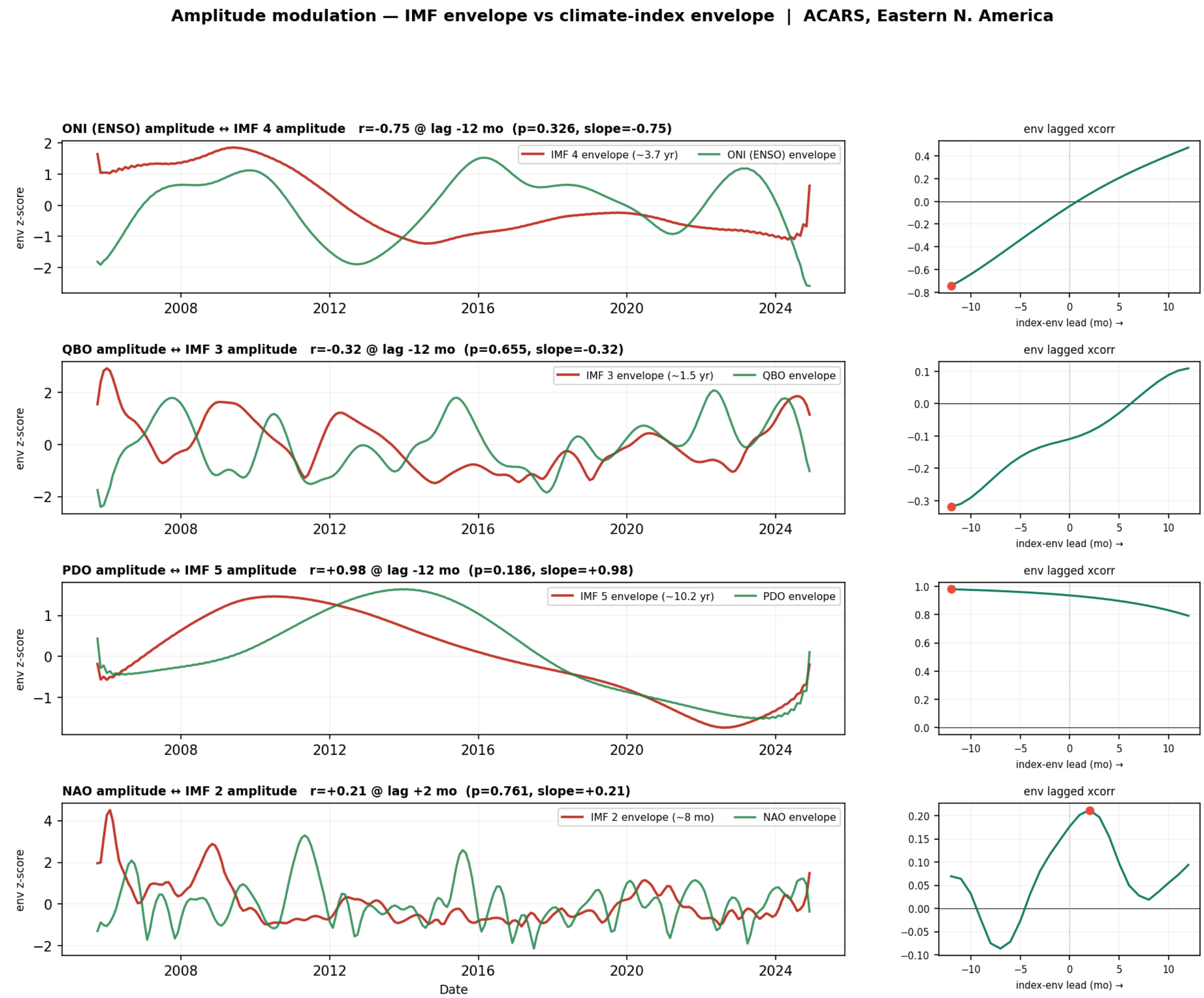

Figure 7: Amplitude-modulation (Hilbert-envelope) test. Apparent envelope correlations (e.g. PDO, $r = +0.98$) are driven by autocorrelation and the small number of multi-year cycles ($p = 0.19$, n.s.), not a real teleconnection.

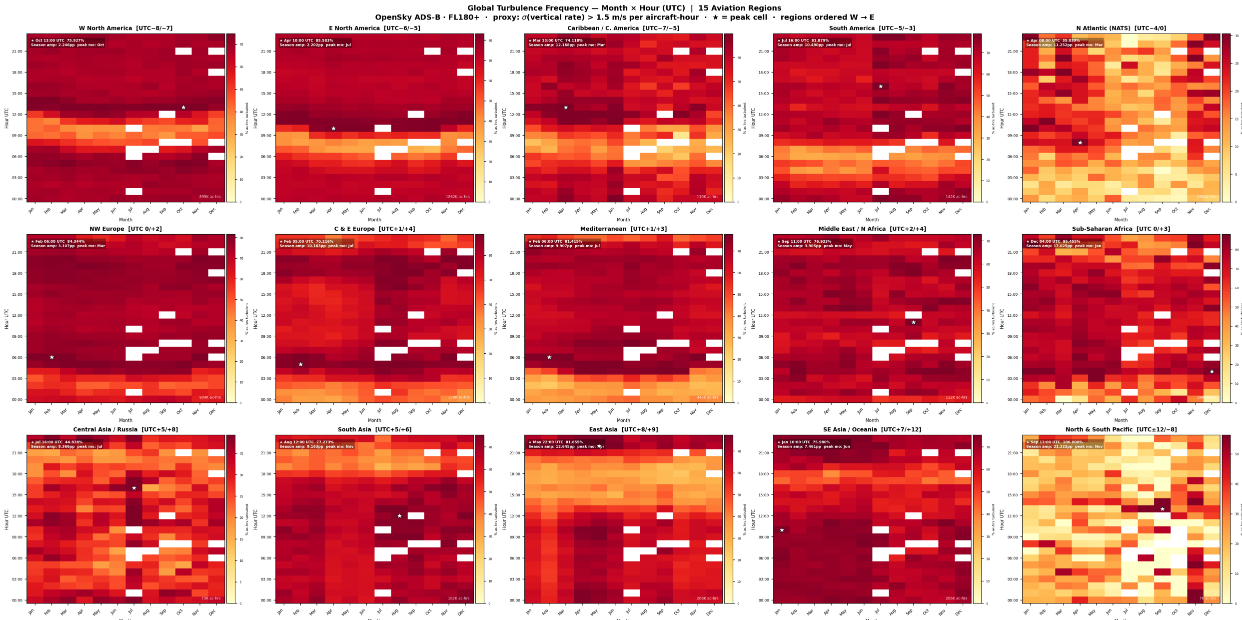

Figure 8: Month × hour (UTC) turbulence-proxy heatmaps for 15 global aviation regions from OpenSky ADS-B (vertical-rate variance proxy, std(w) $> 1.5\,\mathrm{m\,s^{-1}}$). Peak cells ($\star$), seasonal amplitude, and sample sizes are annotated per region; regions below 200 aircraft-hours are flagged insufficient.

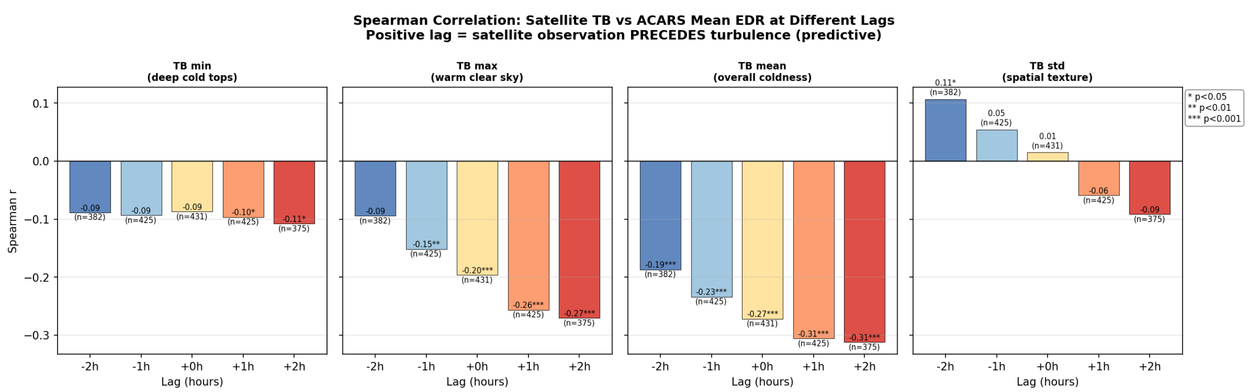

Figure 9: Spearman correlation of satellite brightness-temperature statistics with ACARS mean EDR at lags -2 to $+2\,$h. Positive lag = satellite precedes turbulence; `tb_mean` at $+2\,$h gives $r = -0.31$.

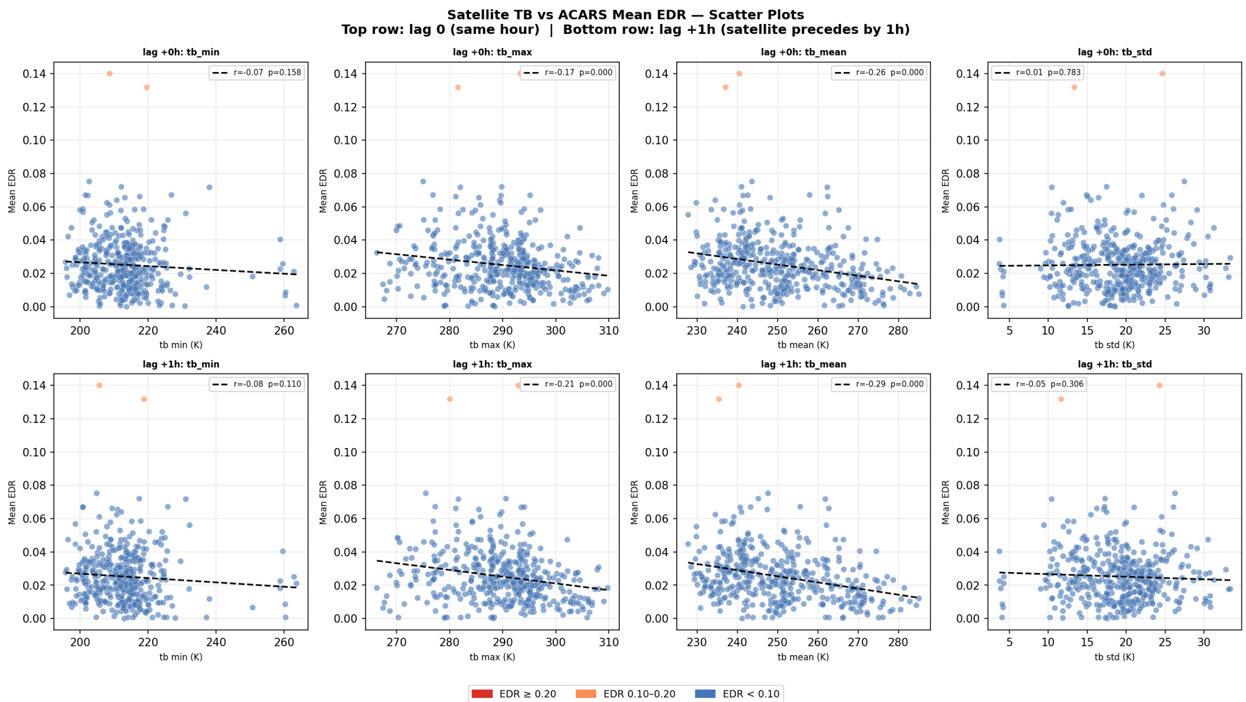

Figure 10: Satellite TB vs. ACARS mean EDR scatter at lag 0 (top) and lag +1 h (bottom). Colder `tb_mean`/`tb_max` associates with higher EDR; severe clear-sky events (no cold cloud) are the off-trend outliers.

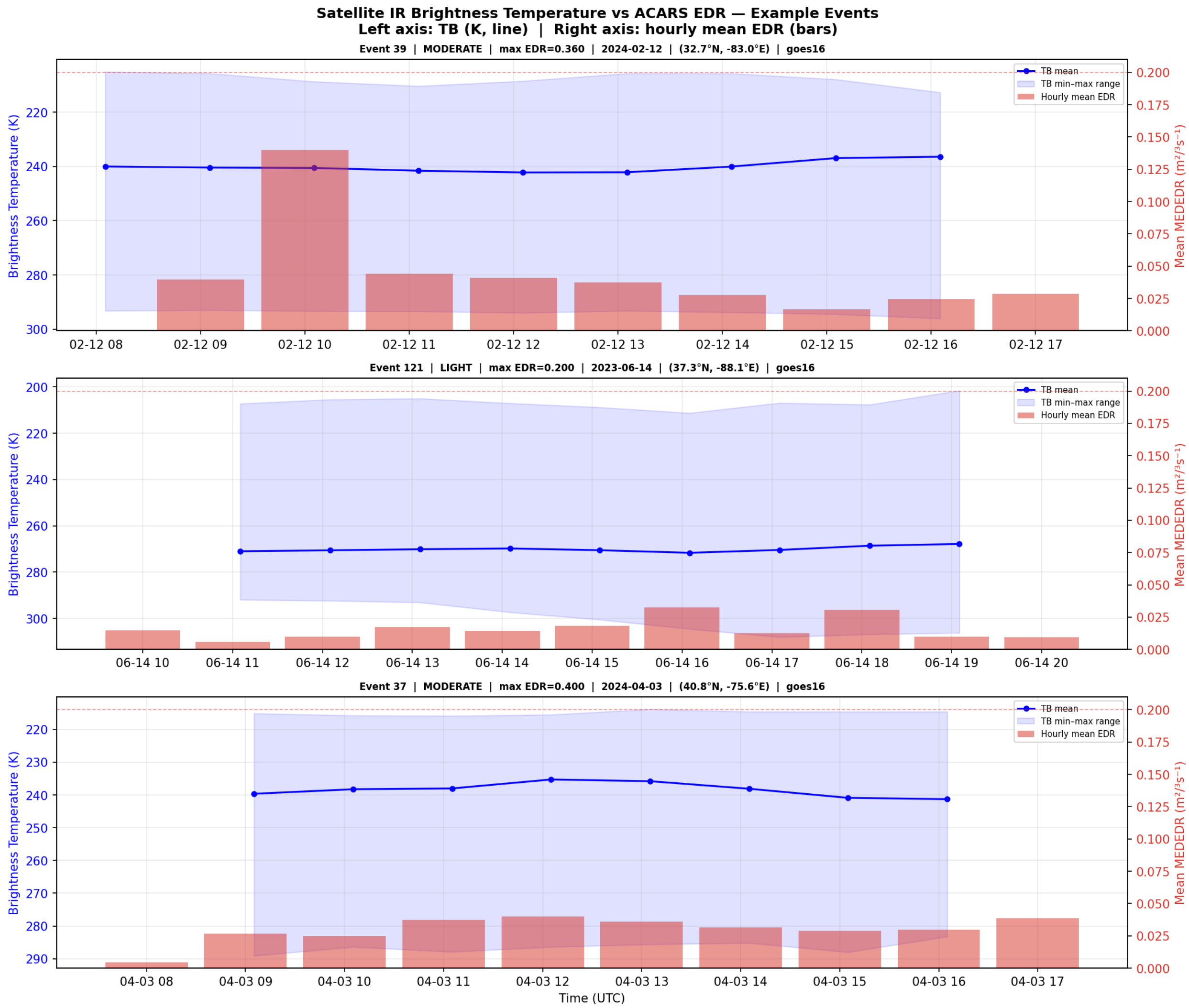

Figure 11: Example events: satellite IR brightness temperature (line) vs. hourly mean EDR (bars). Cloud-top cooling tends to lead EDR peaks in convective cases.

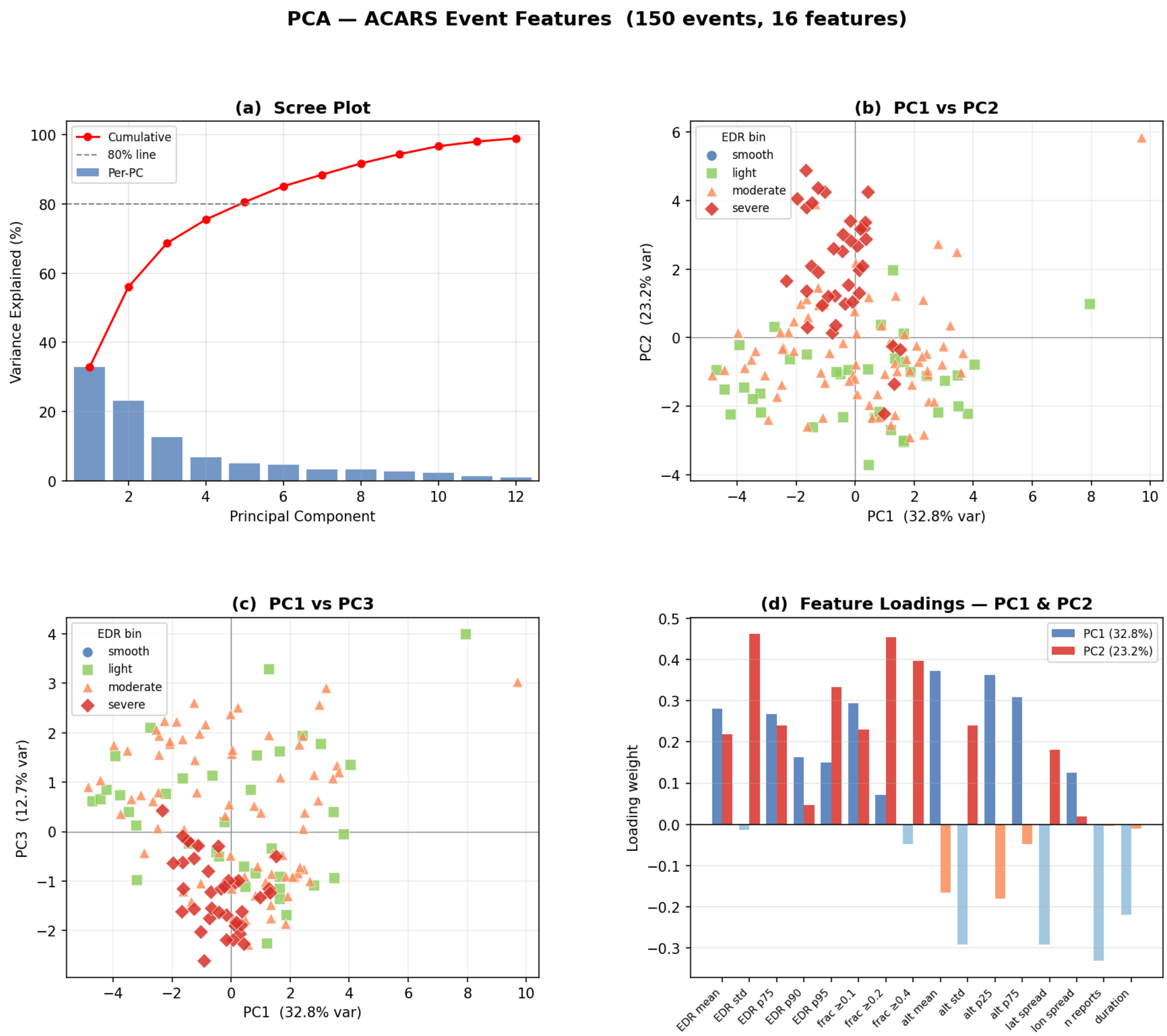

Figure 12: PCA of ACARS event features (150 events, 16 features). PC1 (32.8%) is an EDR-intensity axis along which severe events separate; scree, PC1–PC2, PC1–PC3, and feature loadings shown.

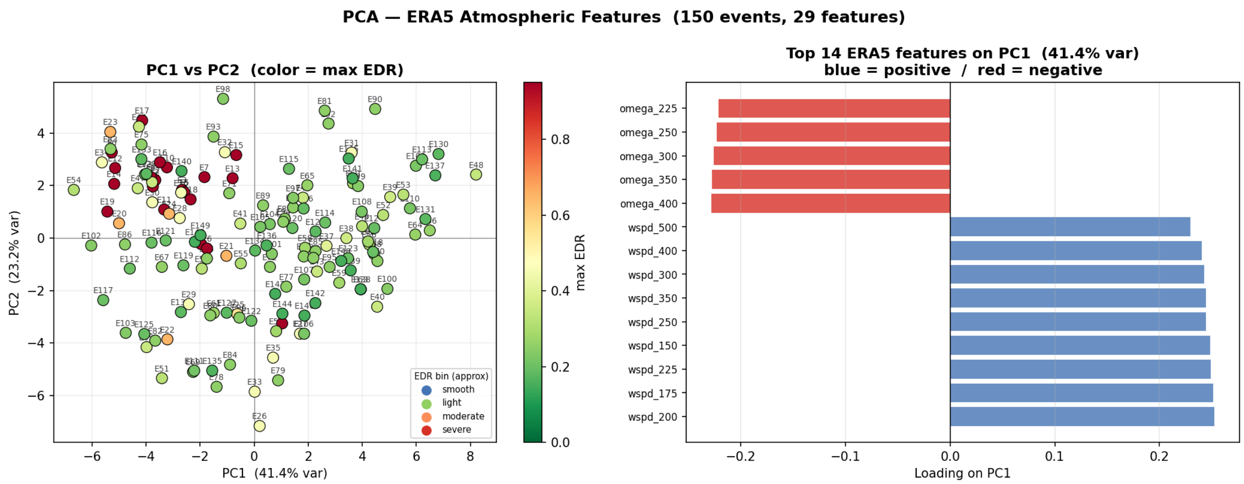

Figure 13: PCA of ERA5 atmospheric features (29 features). PC1 (41.4%) opposes vertical velocity against per-level wind speed (a shear/stability axis), but EDR severity is only weakly separated—motivating a spatial model over pooled features.

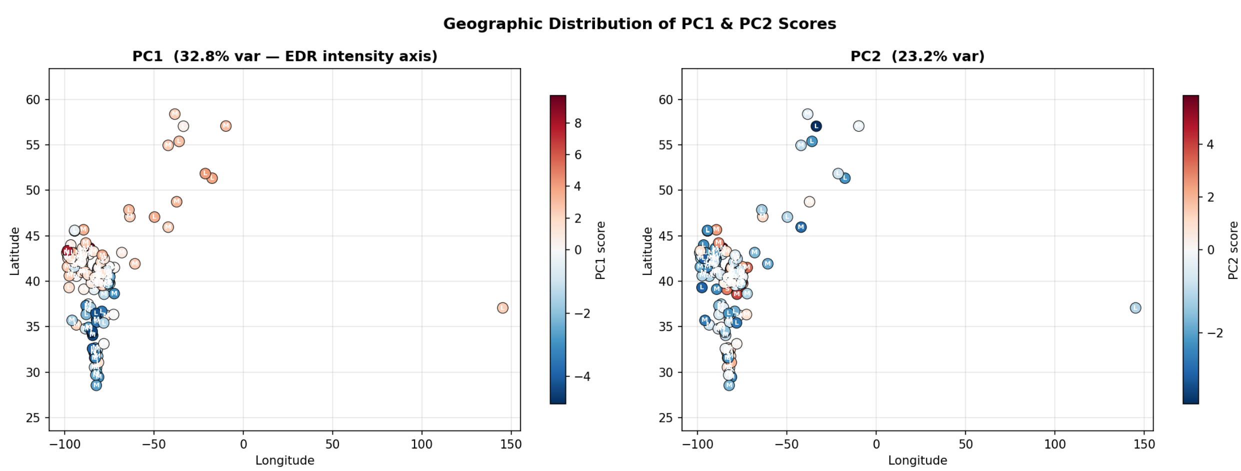

Figure 14: Geographic distribution of PC1 (EDR-intensity) and PC2 scores by event centroid; most intense events cluster at higher latitudes.

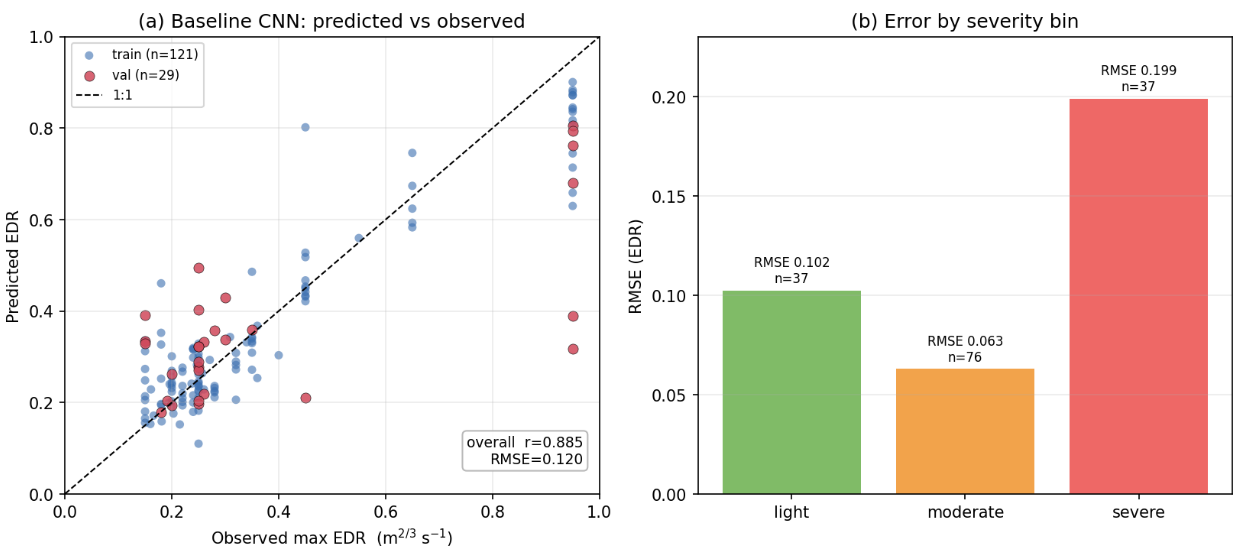

Figure 15: Preliminary NumPy-CNN predictions. (a) Predicted vs. observed max EDR (train blue, validation red, 1:1 dashed); overall $r = 0.885$. (b) RMSE by severity bin—error concentrates in the rare severe tail.